

Number theory – 3

Quadratic residues and reciprocity

Homework

14 сентября 2026 года

Theoretical minimum. Exactly half of the non-zero residues modulo an odd prime are squares. Euler's criterion says $\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}$, whence the Legendre symbol is multiplicative and $\left(\frac{-1}{p}\right) = 1$ exactly for $p \equiv 1 \pmod{4}$. Gauss's lemma expresses the symbol as $(-1)^\mu$, where μ counts the multiples of a landing in the upper half, and gives $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}$. Quadratic reciprocity $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}$ follows from counting lattice points in two triangles tiling a rectangle, and makes every symbol computable in the time of the Euclidean algorithm; the Jacobi symbol extends the computation but not the interpretation. A prime is a sum of two squares exactly when it is 2 or $\equiv 1 \pmod{4}$, and an integer exactly when every prime $\equiv 3 \pmod{4}$ enters to an even power. The quadratic Gauss sum equals $\sqrt{p}$ or $i\sqrt{p}$ and is evaluated in Fourier analysis – 4.

1. Prove that the number of solutions of $x^2 + y^2 \equiv 1 \pmod{p}$ is $p - \left(\frac{-1}{p}\right)$, and more generally count the points of a conic over $\mathbb{F}_p$ by completing the square and summing Legendre symbols.
2. Let q be an odd positive integer and let N_q be the number of integers a with $0 < a < q/4$ and $\gcd(a, q) = 1$. Show that N_q is odd if and only if $q = p^k$ with $k \geq 1$ and p a prime congruent to 5 or 7 modulo 8. (Putnam 2014)
3. Prove Thue's lemma: for p prime and $p \nmid a$ there are x, y with $0 < |x|, |y| < \sqrt{p}$ and $x \equiv ay \pmod{p}$. Deduce the two-squares theorem, and use the same pigeonhole to prove that every prime $p \equiv 1 \pmod{4}$ is of the form $x^2 + y^2$.

4. Prove Legendre's three-square theorem in the easy direction: an integer of the form $4^a(8b + 7)$ is not a sum of three squares. Then prove Lagrange's four-square theorem, using the identity of Euler and the descent argument.
5. Prove that the Jacobi symbol satisfies the same reciprocity law, give an example where $\left(\frac{a}{n}\right) = 1$ although a is not a square modulo n , and explain how the symbol is used in the Solovay–Strassen primality test.
6. Determine all pairs of positive integers (x, y) such that $\sqrt[3]{7x^2 - 13xy + 7y^2} = |x - y| + 1$. (IMO shortlist 2014, N2)
7. Find all triples (p, x, y) with p prime and x, y positive integers such that $x^{p-1} + y$ and $x + y^{p-1}$ are both powers of p . (IMO shortlist 2014, N5)
8. Find the smallest number n such that there exist polynomials $f_1, \dots, f_n$ with rational coefficients satisfying $x^2 + 7 = f_1(x)^2 + \dots + f_n(x)^2$. (IMO shortlist 2010, N3)
9. Let p be an odd prime. Prove that $\sum_{a=1}^{p-1} \left(\frac{a}{p}\right) a$ is divisible by p when $p \equiv 1 \pmod{4}$, and relate the value for $p \equiv 3 \pmod{4}$ to the class number, quoting the classical formula.
10. Prove that the quadratic Gauss sum $g = \sum_{a=0}^{p-1} e^{2\pi i a^2/p}$ satisfies $g^2 = \left(\frac{-1}{p}\right) p$, and deduce quadratic reciprocity from the behaviour of g under raising to the q -th power modulo q . (The evaluation of the sign of g is in Fourier analysis – 4.)